

HUZEFA JAMBUGHODA

huzefa.jambughoda@gmail.com ✉ [GitHub](#) <> [LinkedIn](#)  +971-585960011 📞 Dubai, UAE 📍

AI Engineer with 5+ years of experience bridging ambiguous business problems and production-grade AI. Proven expertise in architecting Agentic workflows, RAG pipelines and scalable ML infrastructure, transitioning from rigorous traditional ML to applied GenAI.

SKILLS

Languages & Backend: Python, C++, SQL, FastAPI, Pydantic

ML/DL: PyTorch, Scikit-Learn, Time-Series Forecasting

GenAI: LangChain, LangGraph, RAG, LLM Gateways, Qdrant, FAISS, Milvus, Ragas, DeepEval

LLM: Finetuning, HuggingFace, QLoRA/PEFT, LLM-as-a-Judge, Guardrails

Infra & MLOps: Kubernetes, Docker, MLFlow, Vertex AI, AWS Lambda, OpenTelemetry

EDUCATION

Indian Institute of Technology, Madras | B.Sc. in Data Science & Programming

2020-23 | CGPA: 8.80

Charotar University of Science and Technology | B.Tech. in Information Technology

2016-20 | CGPA: 9.28

EXPERIENCE

ACDC

Dubai, UAE

Founder & Lead AI Engineer

Sep 2023 - Present

- Architected an Agentic AI product using **ReAct architecture** and **LangGraph**, reducing manual ERP data entry time by 80% by orchestrating a multi-step agent pipeline with **Qdrant** vector database and **Mem0** episodic memory.
- Engineered a centralized **LLM Gateway** with semantic **caching in RAG**, cutting API token costs by 33% and improving response latency, deployed via **Kubernetes** and exposed through **FastAPI** with Pydantic-validated schemas.
- Fine-tuned Kimi K2 via HuggingFace Spaces using **QLoRA/PEFT** on UAE construction industry data, improving domain-specific compliance query accuracy for Leo2.
- Tracked experiments and monitored inference pipelines using MLflow and an OpenTelemetry stack, ensuring high observability and reliability for production agents.
- Led B2B client discovery and handling, translating ambiguous operational needs into concrete technical specs.

Google (via Infocusp)

Ahmedabad, India

Senior Machine Learning Engineer

Dec 2022 - Aug 2023

- Reduced genomic features from 50,000+ to ~100, using Monte Carlo Dropout as well as Gradient Boosting, improving gene edits quality by 30%.
- Accelerated research iteration speed by building an end-to-end ML pipeline that is fully YAML-configured and supports custom NNs in **PyTorch** and all **Scikit-Learn** models with hyper-parameter tuning and automated Docker deployments on **VertexAI**.
- Integrated Monte Carlo Dropout & Variational Dropout to reduce the effects of network sparsity.
- Built **7 data pipelines** using Pydantic and Pandas to transform raw DNA sequences to trainable features.
- Developed a Computer Vision pipeline on Vertex AI with crowd-sourced labelling to automate physical trait collection from live footage, eliminating hours of manual field observation.

Infocusp

Ahmedabad, India

Machine Learning Engineer

Feb 2021 - Dec 2022

- Autodesk: Boosted predictive accuracy from 74% to 87% for chemical dosage using **LSTM**, **LightGBM** and **XGBoost**, reducing raw-material usage and saving operational energy at water treatment plants.
- Bryte: Engineered serverless, event-driven **AWS Lambda** pipelines to preprocess live IoT smart bed data, shifting from 8-hour batch windows to real-time availability.
- Legal Sifter: **Fine-tuned BERT** architectures to optimize embedding strategies for contract clause classification, boosting parsing accuracy by 18%.

Schneider Electric

Bangalore, India

SE Intern

Feb 2020 - May 2020

- Analyzed energy consumption data from active deployments to identify usage patterns and operational anomalies.

- Researched ML approaches for real-time phishing detection, evaluating **feature engineering** strategies on URL structure and webpage contents.

Enlighten Infosystems

SE Intern

Vadodara, India
Apr 2019 - May 2019

- Built an interactive **geospatial visualization** tool in Python and Bokeh.
- Compared evolutionary and swarm algorithms for **feature selection** from scratch on genetic data - Artificial Bee Colony, Ant Colony Optimization and Genetic Algorithm.

Bhaskaracharya Institute of Space Applications and Geo-Informatics

SE Intern

Gandhinagar, India
Apr 2018 - May 2018

- Automated GIS workflows and deployed them as 3 Python **plugins** for on-premise QGIS application.
- Reduced manual trimming work of ~5mins per spatial feature to seconds utilizing convex hull optimization.

PROJECTS

Leo2

- A multi-tenant compliance management platform for the UAE construction industry, enabling Question Answering over long sequence technical documents (RFIs, specs, contracts) shared across multiple organizations via a fine-tuned, domain-expert LLM.
- Finetuned Kimi K2 using QLoRA/PEFT for construction domain document understanding, deployed on HuggingFace Spaces with a RAG layer for clause-level retrieval over proprietary documents.
- Stack: Kimi K2, QLoRA, PEFT, HuggingFace, RAG, LangGraph, Qdrant, FastAPI.

YT Ads Compliance RAG System

- A Retrieval-Augmented Generation pipeline that ingests YouTube videos, extracts multimodal content and generates compliance reports - reducing manual review time by over 80%.
- Designed a LangGraph workflow integrating Azure Video Indexer for multimodal extraction, Azure AI Search for semantic retrieval and Microsoft Azure Foundry for LLM inference.
- Stack: RAG, LangGraph, Azure Video Indexer, Azure AI Search, Microsoft Azure Foundry.

Mandli

- A multi-agent pipeline that researches topics, writes SEO-friendly content and publishes them periodically.
- Stack: CrewAI, LangGraph, GLM-5.2 (orchestrator), MiniMax M3 (research), Claude Sonnet (writing), Kimi K2.6 (seo), FireCrawl MCP (live web ingestion), Qdrant, FastAPI, Astro JS (frontend), Streamlit (dashboard).

OpenAccounts

openaccounts.vercel.app

- Fully client-side, minimal, double-entry accounting app with statement ingestion using dual-stage doc parsing.
- Stack: React, Tailwind, IndexedDB, Google Drive, OAuth2, PyMuPDF, PPDLayout v3, GLM-OCR.

CERTIFICATIONS & ACHIEVEMENTS

Building with Claude API - Anthropic Academy	Certified
AI Agent - HuggingFace	Certified
AI Engineer with Foundry and AIP - Palantir Learn	Certified
National Creativity & Aptitude Test - IFEHE India	National #9
Programming, Data Structures & Algorithms using Python - NPTEL	Top 1%
Codechef	huzefa_j - 5 star